

Statistical Inference & Hypothesis Testing Report

Analyst: Portfolio Toolkit | Date: June 02, 2026 | Framework: AP Statistics Standard

1. State

Define the hypotheses and significance level to evaluate the claims of the A/B test.

Significance Level (α): 0.05

Hypotheses (Symbols):

$$H_0 : \mu_A = \mu_B \quad H_a : \mu_A \neq \mu_B$$

Null Hypothesis (H_0): There is no difference in the means of the metrics between Group A and Group B ($\mu_A = \mu_B$).

Alternative Hypothesis (H_a): There is a statistically significant difference in the means of the metrics between Group A and Group B ($\mu_A \neq \mu_B$).

2. Plan

Identify the appropriate statistical procedure and verify its mathematical assumptions.

Recommended Procedure: Welch's T-Test (Unequal Variances)

Assumption / Condition Checked	Status & Diagnostic Details
Randomness	Assuming independent random sampling/assignment of groups.
Independence	Samples are mutually independent and represent < 10% of population.
Normality (Shapiro-Wilk)	Group A Normality (p = 0.7219, Passed). Group B Normality (p = 0.8531, Passed). Normality met.
Equal Variance (Levene's)	Levene's Test for equal variance: p = 0.0150 (Unequal variances assumed).

■ CONDITIONS MET

All mathematical conditions for the Welch's T-Test (Unequal Variances) are satisfied. Parametric results are highly reliable.

3. Do

Calculate sample metrics, standard error, test statistic, and the resulting p-value.

Sample Descriptive Statistics:

Group	Sample Size (n)	Sample Mean (x̄)	Standard Deviation (s)	Variance (s²)
Group A	50	45.2701	9.2023	84.6828
Group B	55	54.6722	13.3346	177.8121

Calculations & Mathematical Formulations:

• Se:

$$SE = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = 2.2196$$

• T Stat:

$$t = \frac{\bar{x}_B - \bar{x}_A}{SE} = 4.2360$$

• Df:

$$df = 96.28$$

• Ci:

$$CI = (\bar{x}_B - \bar{x}_A) \pm t_{crit} \times SE = (4.9964, 13.8078)$$

Statistical Parameter	Value
Test Statistic (t)	4.23595
p-value	5.21118e-05
Degrees of Freedom (df)	96.28
95% Confidence Interval	(4.9964, 13.8078)

4. Conclude

State the final decision and offer a business interpretation based on the statistical results.

Compare p-value to α: 0.00005 < 0.05. Since the p-value is less than the significance level, we **reject the null hypothesis (H₀)**. There is strong evidence supporting a statistically significant difference between the two variations.

■ **DECISION: REJECT NULL HYPOTHESIS**

Business Action Recommendation: The metric difference of 9.4021 is statistically significant. Variation **Group B** outperforms the other and should be adopted to maximize efficiency/revenue.